RDS203_HW2

Tzu-Hsi (Cece) Jen

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```r
library(ggplot2)
library(tidyr)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v lubridate 1.9.5     v tibble    3.3.1
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(gt)
```

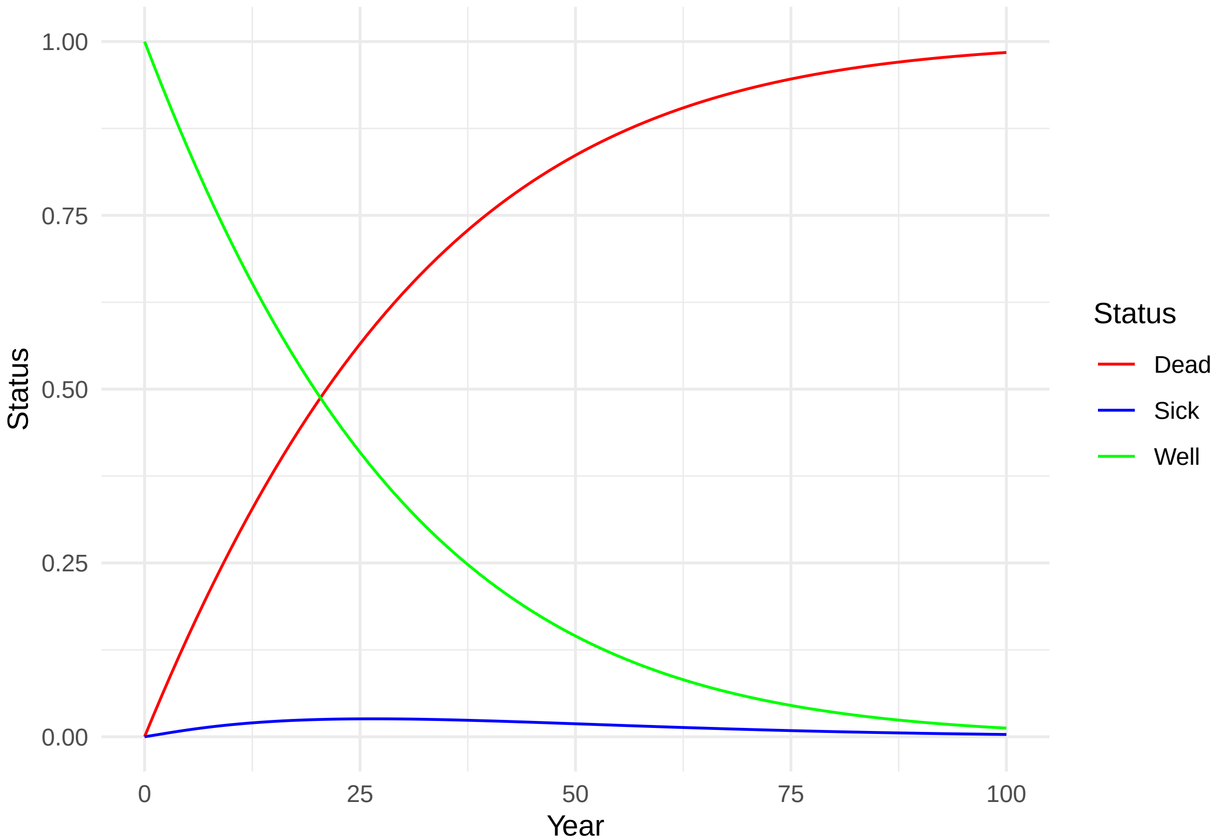
Question 1

1a

```r
data1a <- read.csv("./rds203_problemset2/data/Q1a.csv")
data1a <- data1a|>
  pivot_longer(cols = c("Well", "Sick", "Dead"),
               names_to = "Status",
               values_to = "Proportion")
head(data1a, 3)
```

```
## # A tibble: 3 x 3
##    Year Status Proportion
##   <int> <chr>       <dbl>
## 1     0 Well            1
## 2     0 Sick            0
## 3     0 Dead            0
```

```r
ggplot(data1a, aes(x = Year, y = Proportion, colour = Status))+
  geom_line()+
  scale_colour_manual(values = c("Well" = "green",
                                 "Sick" = "blue",
                                 "Dead" = "red"))+
  labs(x = "Year",
       y = "Status")+
  theme_minimal()
```

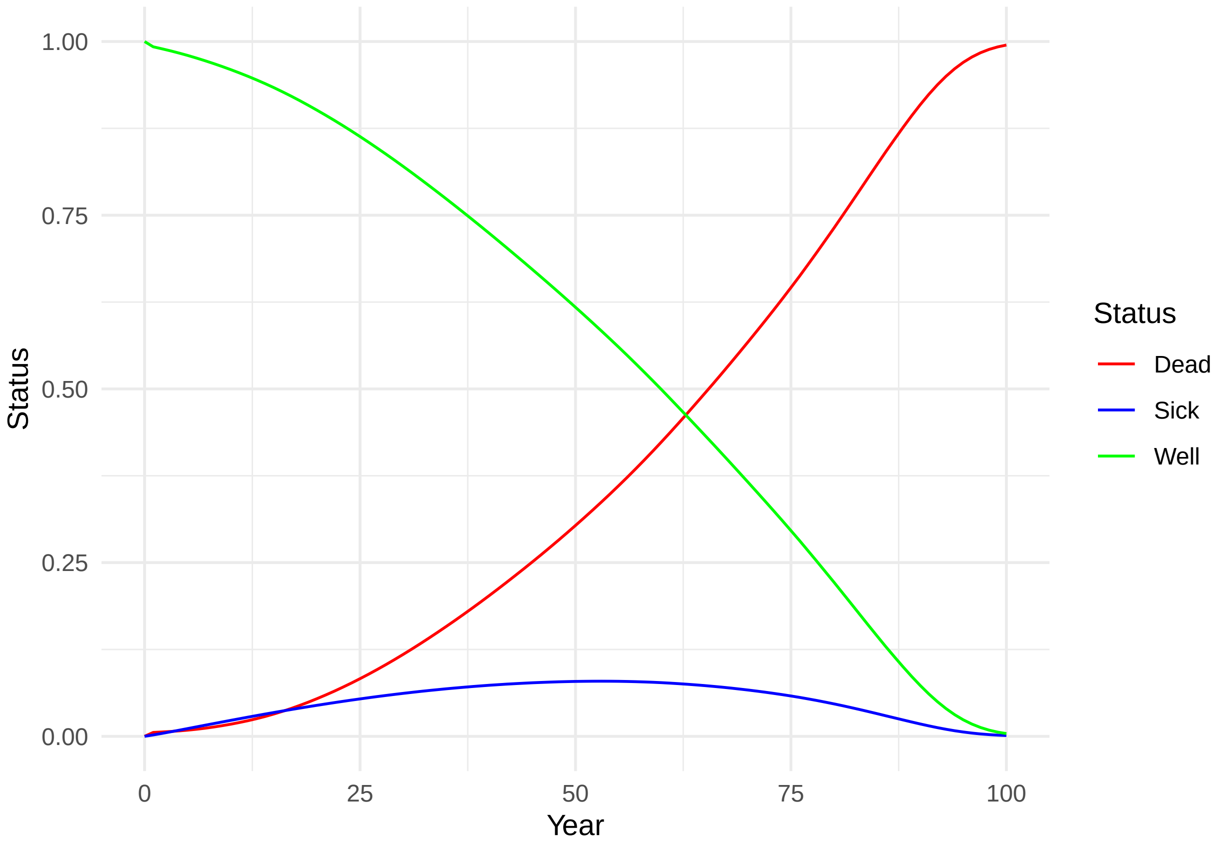


1b

```
data1b <- read.csv("./rds203_problemset2/data/Q1b.csv")
data1b <- data1b|>
  pivot_longer(cols = c("Well", "Sick", "Dead"),
               names_to = "Status",
               values_to = "Proportion")
head(data1b, 3)
```

```
## # A tibble: 3 x 3
##    Year Status Proportion
##   <int> <chr>       <dbl>
## 1     0 Well            1
## 2     0 Sick            0
## 3     0 Dead            0
```

```
ggplot(data1b, aes(x = Year, y = Proportion, colour = Status))+
  geom_line()+
  scale_colour_manual(values = c("Well" = "green",
                                 "Sick" = "blue",
                                 "Dead" = "red"))+
  labs(x = "Year",
       y = "Status")+
  theme_minimal()
```



Question 2

```r
# data2 <- read.csv("./rds203_problemset2/data/Q2.csv")
# gt(data2)
```

Question 3

3e

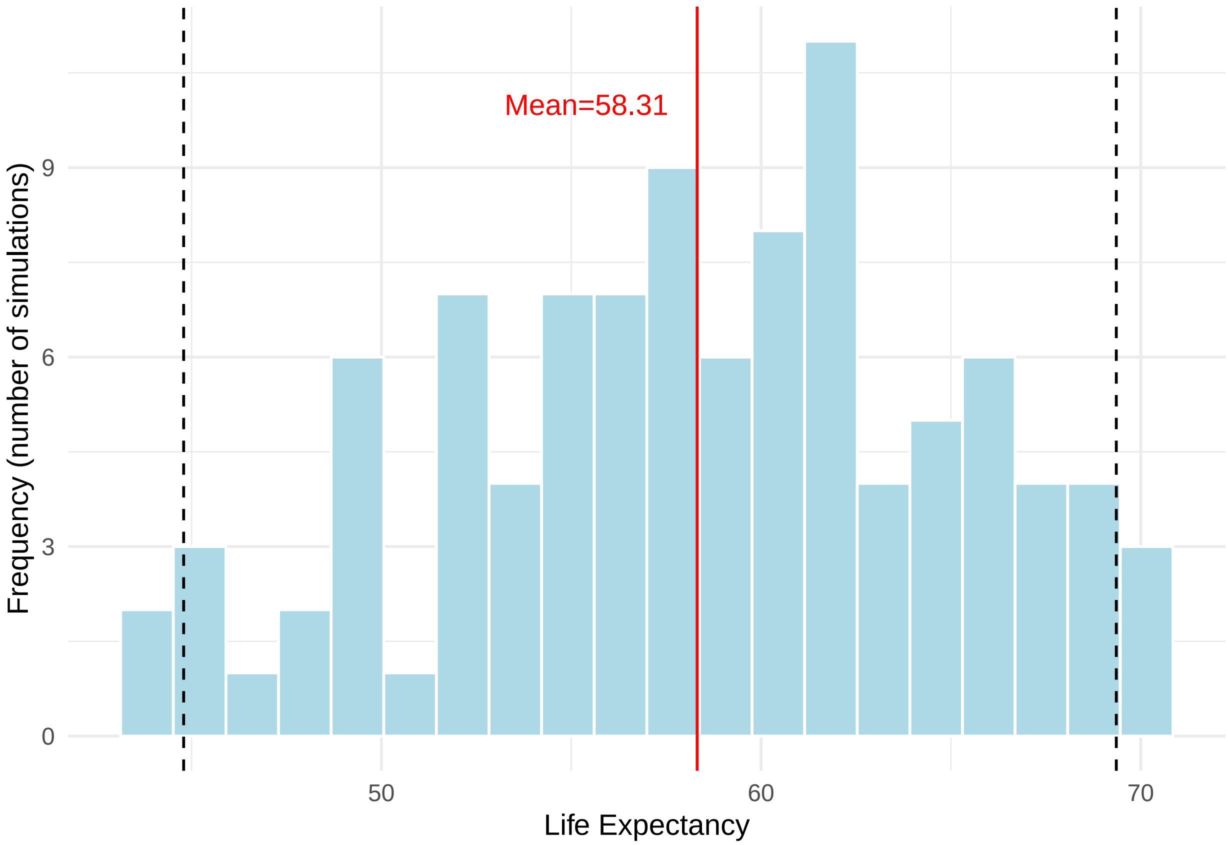
```r
data3e <- read.csv("./rds203_problemset2/data/Q3e.csv")
head(data3e, 3)
```

```
##   LifeExpectancy
## 1       52.65554
## 2       61.42615
## 3       56.85460
```

```r
# Compute mean and 95% uncertainty intervals for plotting
mean_3e <- mean(data3e$LifeExpectancy)
ui_3e <- quantile(data3e$LifeExpectancy, c(0.025, 0.975))

ggplot(data3e, aes(x = LifeExpectancy))+
  geom_histogram(bins = 20, fill = "lightblue", colour = "white")+
  geom_vline(xintercept = mean_3e, colour = "red")+
  geom_vline(xintercept = ui_3e, colour = "black", linetype = "dashed")+
  # Add annotation
  annotate("text",
```

```r
         x = (mean_3e*0.95), y = 10,
         label = paste0("Mean=", round(mean_3e, 2)),
         colour = "red")+
  labs(x = "Life Expectancy",
       y = "Frequency (number of simulations)")+
  theme_minimal()
```




```r
  # theme(
  #   panel.grid.major = element_line(colour = "grey95"),
  #   panel.grid.minor = element_blank()
  # )
```

Question 5

5e

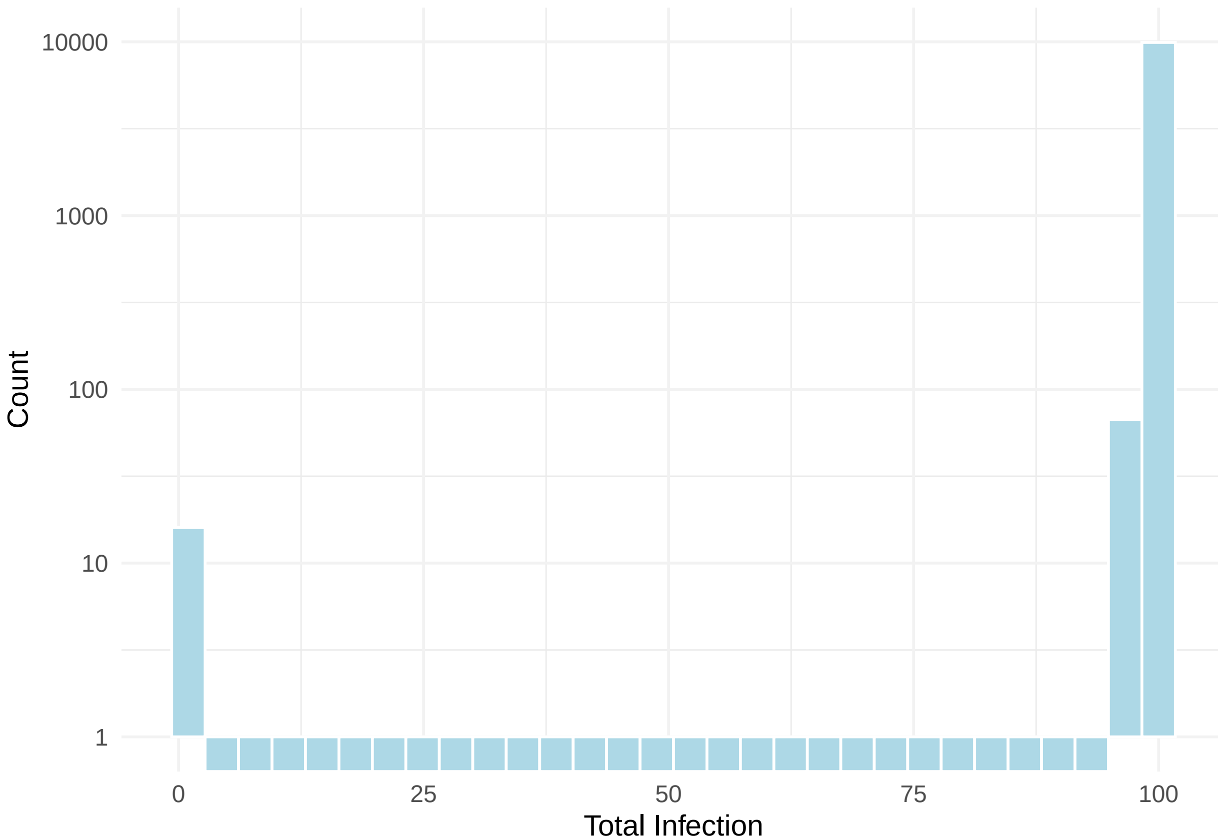
```r
data5e <- read.csv("./rds203_problemset2/data/Q5e.csv")
head(data5e, 3)
```

```
##   TotalInfection Duration
## 1            100       19
## 2             98       20
## 3             99       21
```

```r
# Plot: distribution of the total number of infections
ggplot(data5e, aes(x = TotalInfection))+
  geom_histogram(fill = "lightblue", colour = "white", bins = 30)+
  labs(x = "Total Infection",
```

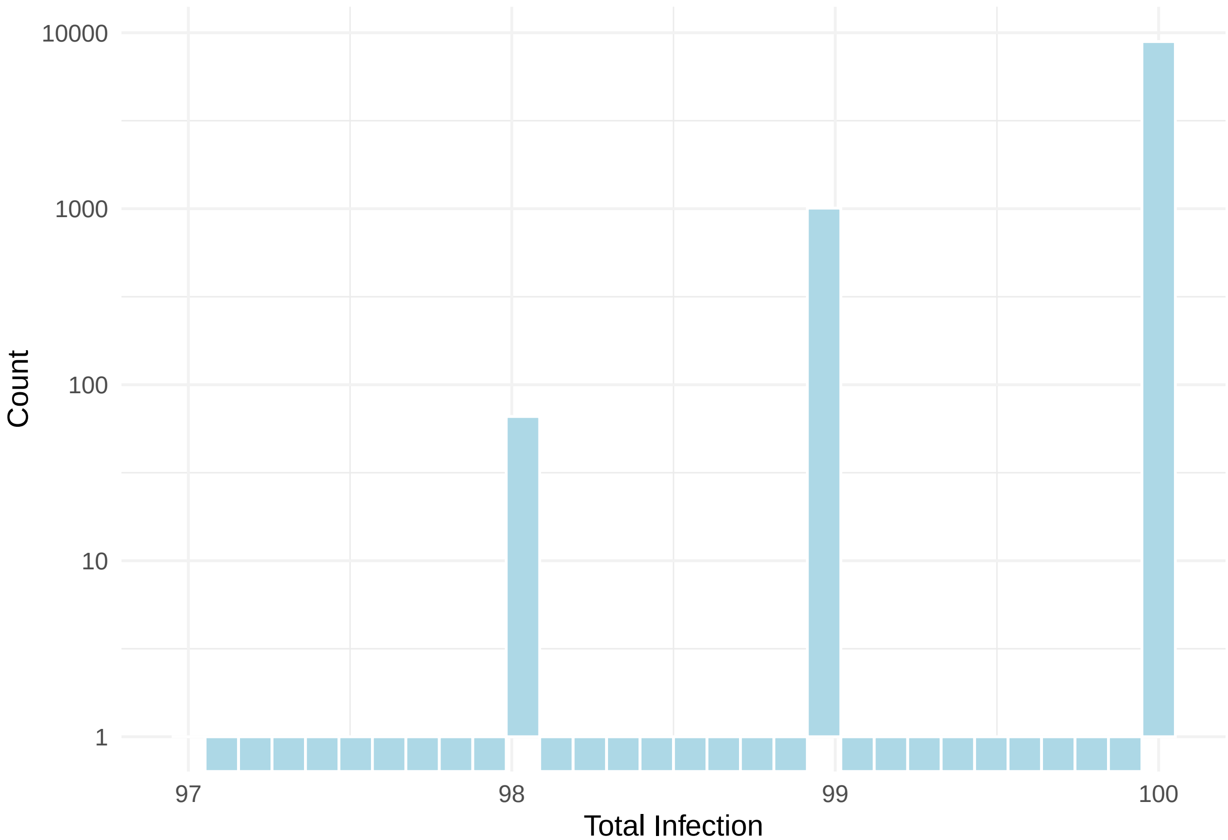
```r
      y = "Count")+
  scale_y_log10()+
  theme_minimal()+
  theme(
    panel.grid.major = element_line(colour = "grey95")
  )
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```

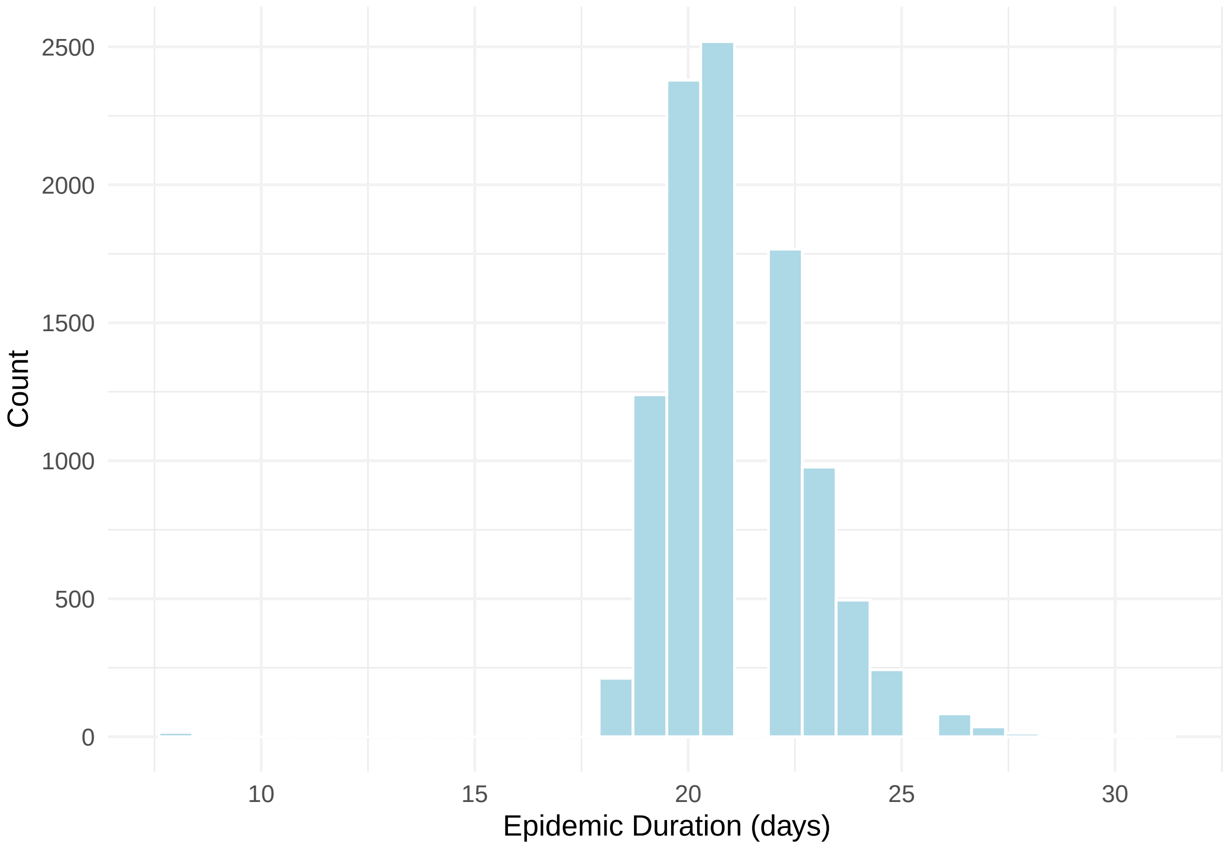



```r
# Plot: distribution of the total number of infections, excluding no-spread cases (TotalInfection == 1)
data5e|>
  filter(TotalInfection > 1)|>
  ggplot(aes(x = TotalInfection))+
  geom_histogram(fill = "lightblue", colour = "white",bins = 30)+
  scale_y_log10()+
  labs(x = "Total Infection",
      y = "Count")+
  theme_minimal()+
  theme(
    panel.grid.major = element_line(colour = "grey95")
  )
```

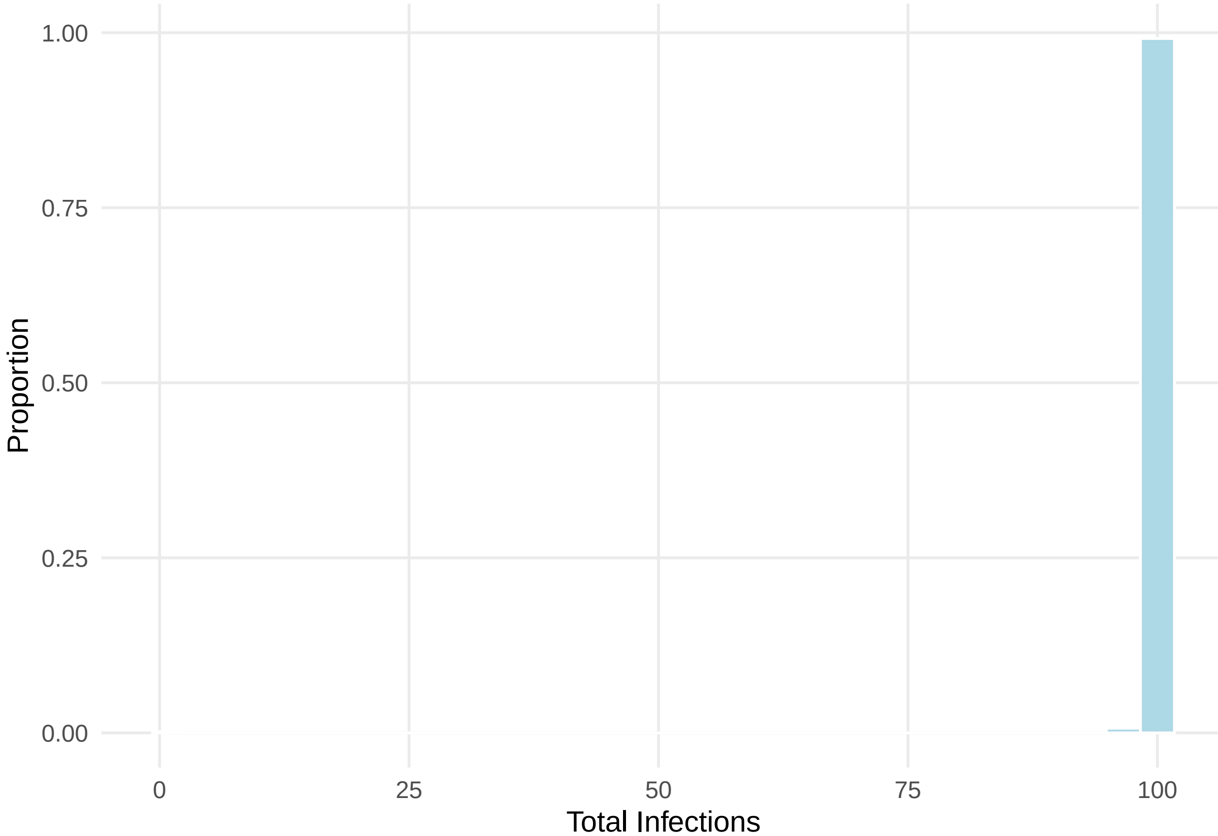
```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```



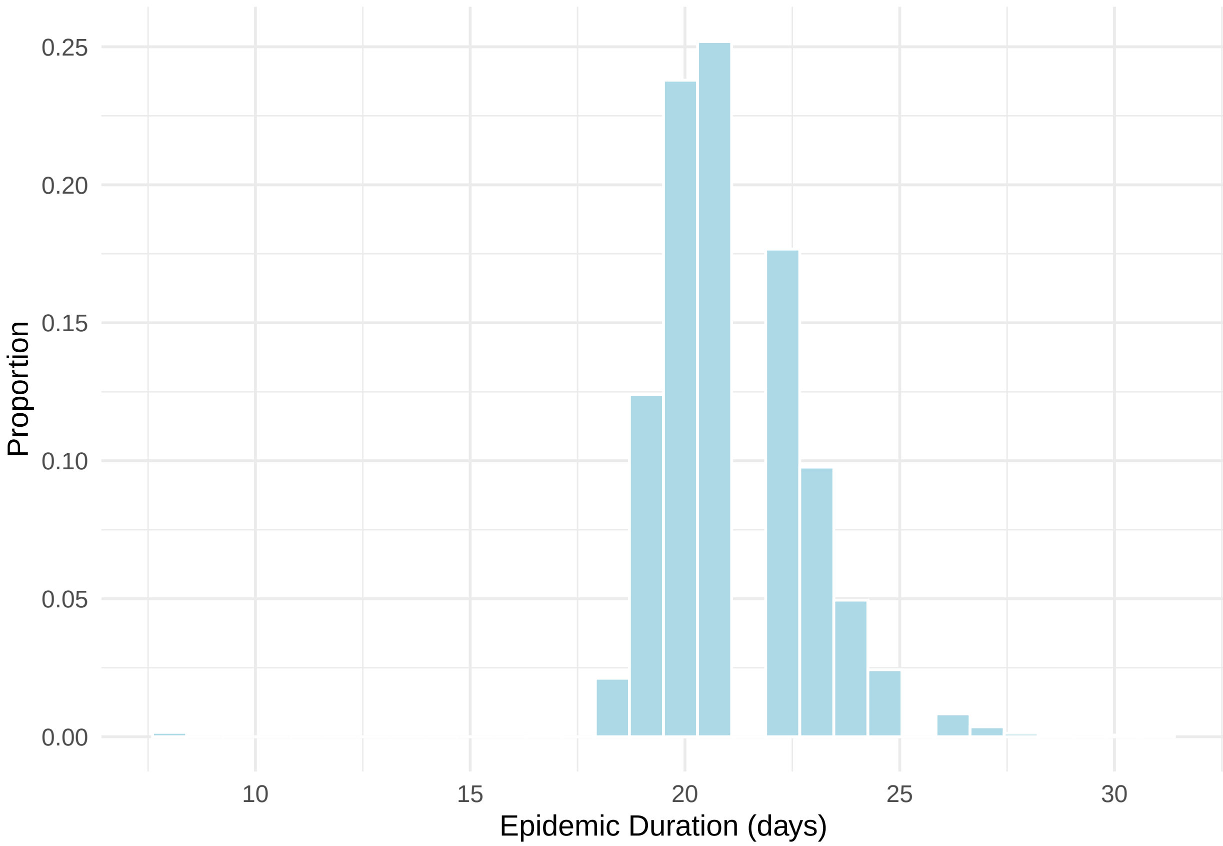

```r
# Plot: length of the epidemic
ggplot(data5e, aes(x = Duration))+
  geom_histogram(fill = "lightblue", colour = "white",bins = 30)+
  labs(x = "Epidemic Duration (days)",
       y = "Count")+
  theme_minimal()+
  theme(
    panel.grid.major = element_line(colour = "grey95")
  )
```




```
ggplot(data5e, aes(x=TotalInfection))+
  geom_histogram(aes(y = after_stat(count / sum(count))),
                   bins = 30,
                   fill = "lightblue",
                   colour = "white")+
  labs(x = "Total Infections",
       y = "Proportion")+
  theme_minimal()+
  theme(panel.grid.minor = element_blank())
```




```r
ggplot(data5e, aes(x=Duration))+
  geom_histogram(aes(y = after_stat(count / sum(count))),
                    bins = 30,
                    fill = "lightblue",
                    colour = "white")+
  labs(x = "Epidemic Duration (days)",
       y = "Proportion")+
  theme_minimal()
```



Question 6

6b

```r
data6a <- read.csv("./rds203_problemset2/data/Q6a.csv") # baseline
data6b <- read.csv("./rds203_problemset2/data/Q6b.csv") # intervention

head(data6a, 2)
```

```
##   LifeExpectancy
## 1        57.4046
## 2        57.7810
```

```r
head(data6b, 2)
```

```
##   LifeExpectancy
## 1        62.5842
## 2        62.6464
```

```r
# Compute mean gain and 95% UI
gain <- data6b$LifeExpectancy - data6a$LifeExpectancy
gainMean <- mean(gain)
ui <- quantile(gain, c(0.025, 0.975))
lower_ui <- ui[1]
upper_ui <- ui[2]

print(paste0("Mean gain: ", round(gainMean, 2)))
```

```
## [1] "Mean gain: 4.98"
```

```r
print(paste0("95% UI: [", round(lower_ui, 2), ", ", round(upper_ui, 2) ,"]"))
```

```
## [1] "95% UI: [4.5, 5.49]"
```

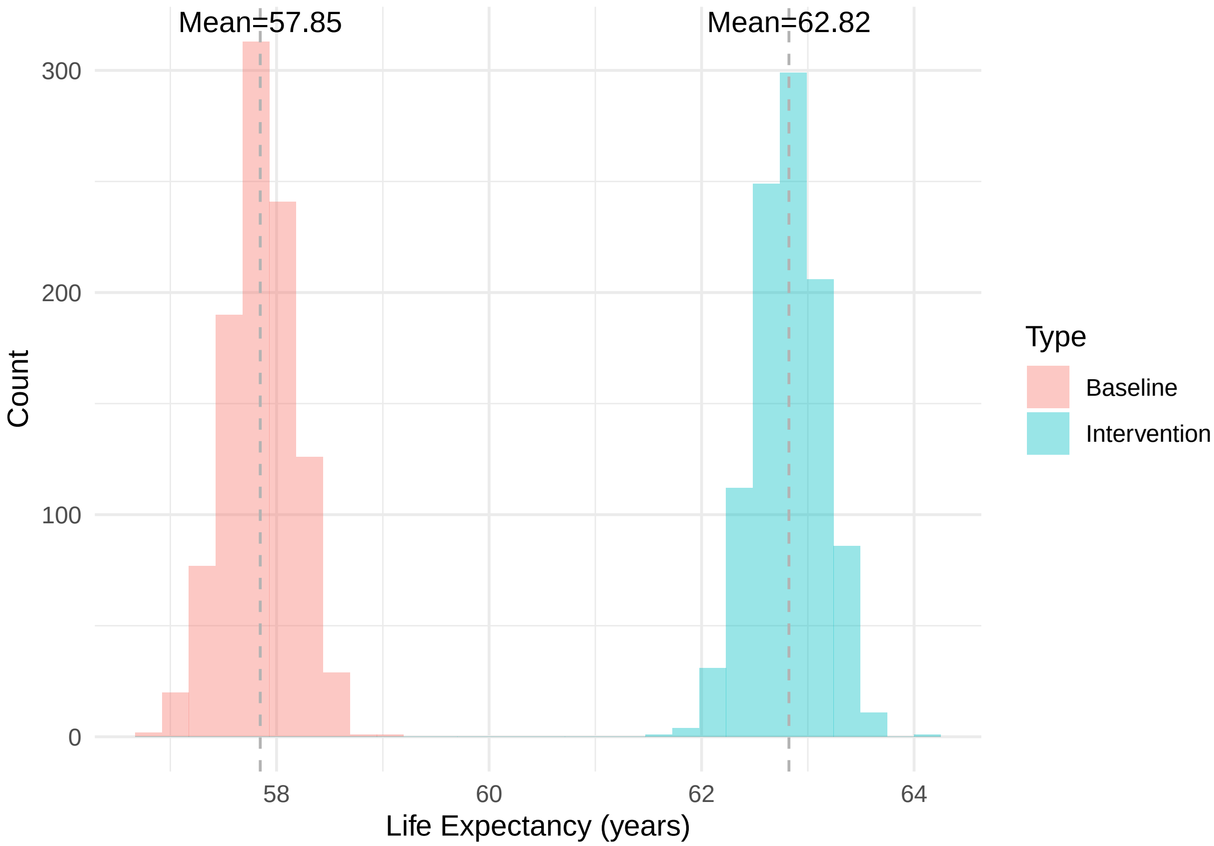
```r
# Plot baseline vs intervention
data6ab_long <- data6a|>
  rownames_to_column("Iteration")|>
  dplyr::rename("Baseline" = "LifeExpectancy")|>
  left_join(data6b|>
              rownames_to_column("Iteration")|>
              dplyr::rename("Intervention" = "LifeExpectancy"),
            by = "Iteration"
            )|>
  pivot_longer(cols = c("Baseline", "Intervention"),
               names_to = "Type",
               values_to = "LifeExpectancy")
head(data6ab_long, 2)
```

```
## # A tibble: 2 x 3
##   Iteration Type         LifeExpectancy
##   <chr>     <chr>                 <dbl>
## 1 1         Baseline               57.4
## 2 1         Intervention           62.6
```

```r
# Compute mean for both groups
mean_df <- data6ab_long|>
  group_by(Type)|>
  summarise(mean_le = mean(LifeExpectancy))

# Plot distribution with mean lines
ggplot(data6ab_long, aes(x = LifeExpectancy, fill = Type))+
  geom_histogram(alpha = 0.4, position = "identity", bins = 30)+

  geom_vline(data = mean_df,
             aes(xintercept = mean_le),
             colour = "grey70", linetype = "dashed")+
  geom_text(data = mean_df,
            aes(x = mean_le, y = Inf,
                label = paste0("Mean=", round(mean_le, 2))),
            vjust = 1.2,
            colour = "black")+
  labs(x = "Life Expectancy (years)",
       y = "Count")+
  theme_minimal()
```

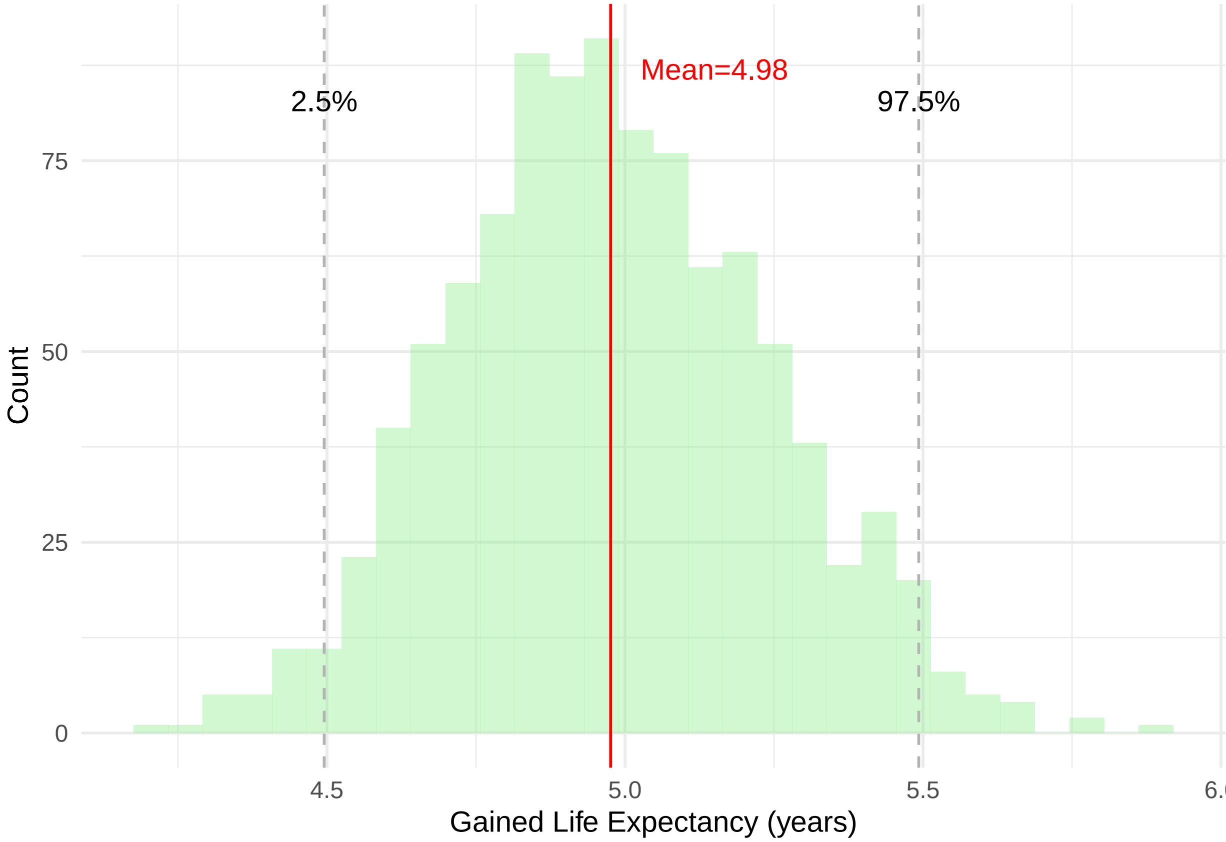



```
# Plot intervention - baseline
data6ab_wide <- data6a|>
  rownames_to_column("Iteration")|>
  dplyr::rename("Baseline" = "LifeExpectancy")|>
  left_join(data6b|>
              rownames_to_column("Iteration")|>
              dplyr::rename("Intervention" = "LifeExpectancy"),
            by = "Iteration"
            )|>
  mutate(gain = Intervention - Baseline)

ggplot(data6ab_wide, aes(x = gain))+
  geom_histogram(fill = "lightgreen", alpha = 0.4, bins = 30)+
  geom_vline(xintercept = gainMean, colour = "red")+
  geom_vline(xintercept = ui, colour = "grey70", linetype = "dashed")+

  annotate("text",
           x = 5.15, y = 87,
           label = paste0("Mean=", round(gainMean, 2)),
           colour = "red")+
  annotate("text",
           x = lower_ui, y = 87,
           label = paste0("2.5%"), vjust = 2)+
  annotate("text",
           x = upper_ui, y = 87,
           label = paste0("97.5%"), vjust = 2)+
```

```r
  labs(x = "Gained Life Expectancy (years)",
       y = "Count")+
  theme_minimal()
```



6c

```r
data6c <- read.csv("./rds203_problemset2/data/Q6c.csv")
head(data6c, 2)
```

```
##   Baseline Intervention   Gain
## 1  57.7800      62.5620 4.7820
## 2  57.7768      62.6696 4.8928
```

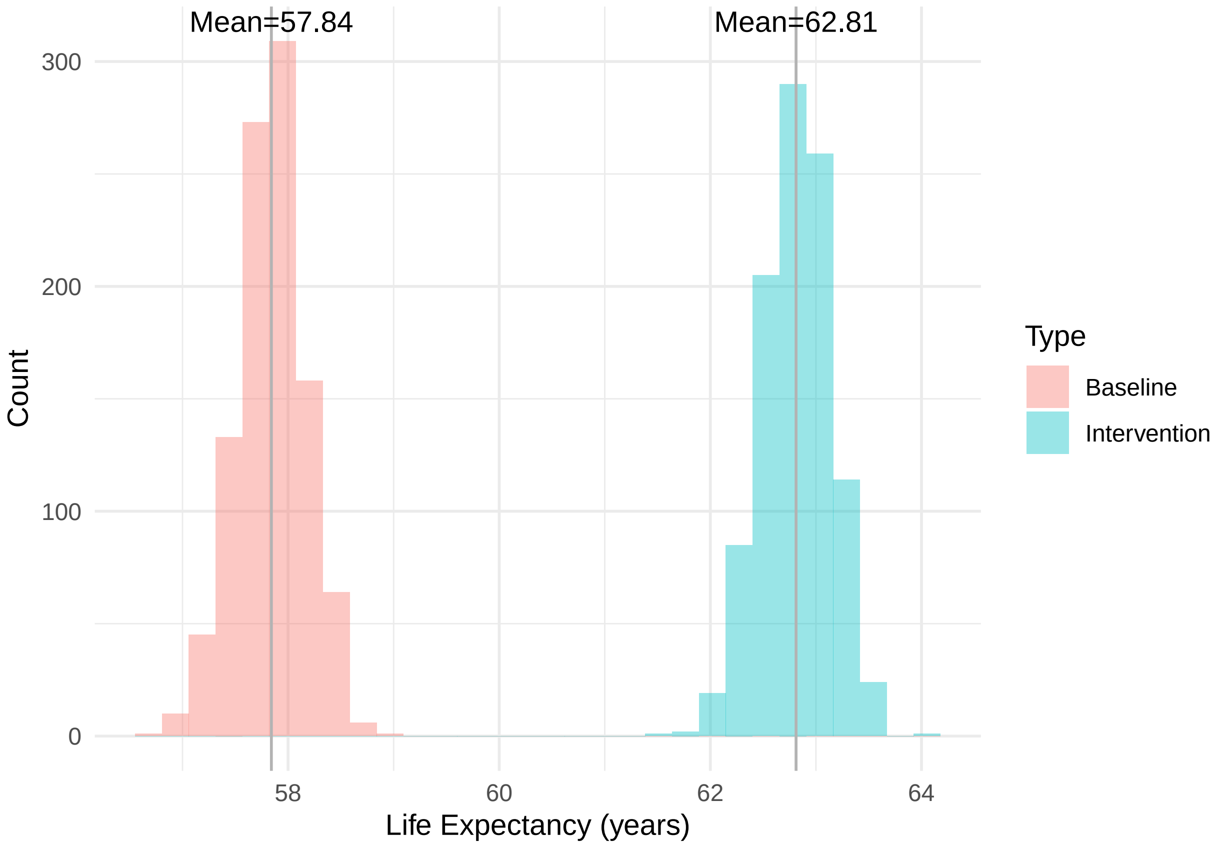
```r
# Plot baseline vs intervention
data6c_long <- data6c|>
  pivot_longer(cols = c("Baseline", "Intervention"),
               names_to = "Type",
               values_to = "LifeExpectancy")

mean_df <- data6c_long|>
  group_by(Type)|>
  summarise(mean_le = mean(LifeExpectancy))

ggplot(data6c_long, aes(x = LifeExpectancy, fill = Type))+
  geom_histogram(alpha = 0.4, position = "identity", bins = 30)+

  geom_vline(data = mean_df,
```

```
             aes(xintercept = mean_le),
             colour = "grey70")+
  geom_text(data = mean_df,
            aes(x = mean_le, y = Inf,
                label = paste0("Mean=", round(mean_le, 2))),
            vjust = 1.2,
            colour = "black")+
  labs(x = "Life Expectancy (years)",
       y = "Count")+
  theme_minimal()
```



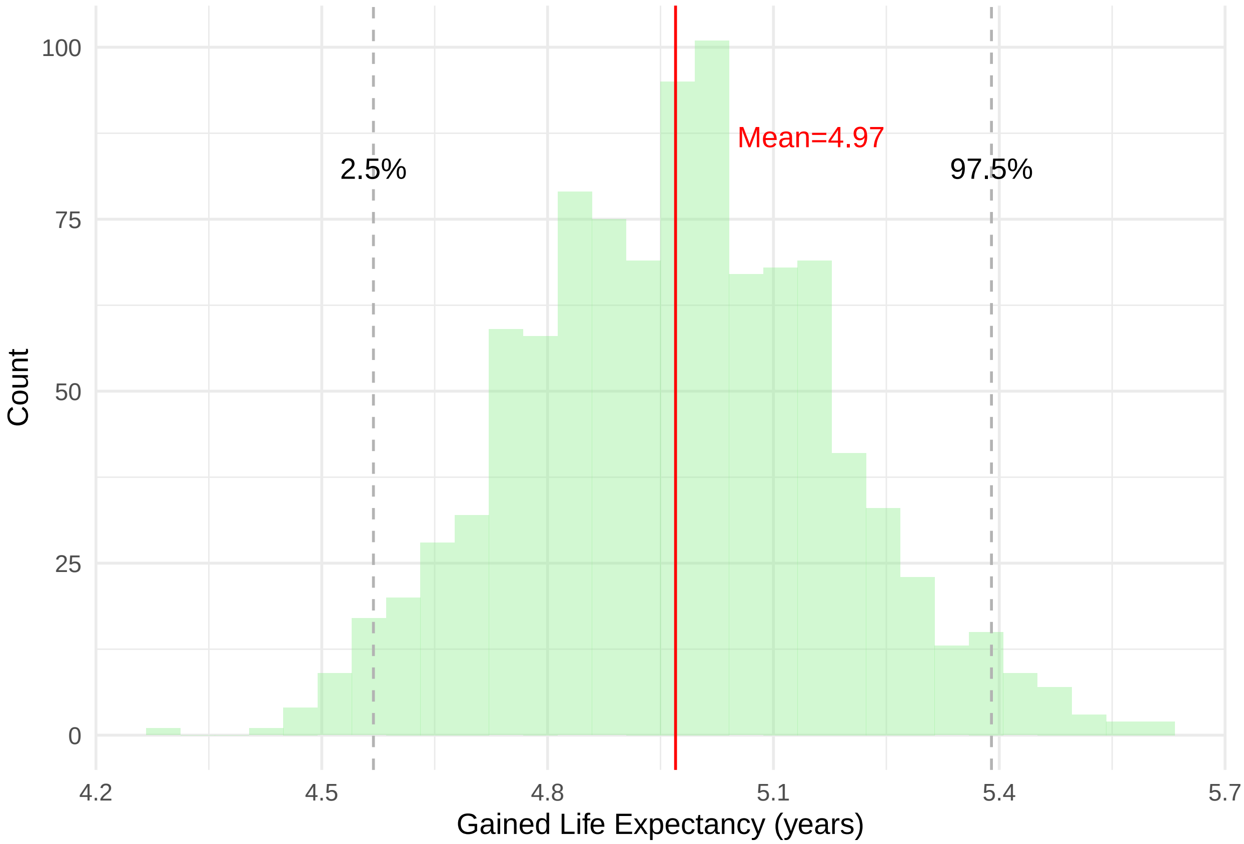

```
# Plot gain
gainMean <- round(mean(data6c$Gain), 2)
ui <- quantile(data6c$Gain, c(0.025, 0.975))
lower_ui <- ui[1]
upper_ui <- ui[2]

ggplot(data6c, aes(x = Gain))+
  geom_histogram(fill = "lightgreen", alpha = 0.4, bins = 30)+
  geom_vline(xintercept = gainMean, colour = "red")+
  geom_vline(xintercept = ui, colour = "grey70", linetype = "dashed")+

  annotate("text",
           x = 5.15, y = 87,
           label = paste0("Mean=", round(gainMean, 2)),
           colour = "red")+
  annotate("text",
```

```r
          x = lower_ui, y = 87,
          label = paste0("2.5%"), vjust = 2)+
  annotate("text",
          x = upper_ui, y = 87,
          label = paste0("97.5%"), vjust = 2)+

  labs(x = "Gained Life Expectancy (years)",
       y = "Count")+
  theme_minimal()
```



Question 7

7e

```r
# ---------------
# Load datasets
# ---------------
# Simulated
data7e <- read.csv("./rds203_problemset2/data/Q7e.csv")
data7e
```

```
##         beta0        beta1  recovery    mortDx   prev_0   prev_1   prev_2
## 1 0.01210111 0.0008811535 0.3573952 0.1462546 5.395328 7.81328 9.568771
##     prev_3   prev_4   prev_5   mort_0   mort_1   mort_2   mort_3   mort_4
## 1 12.38986 13.37023 15.92211 10.77236 11.02236 12.51841 9.288824 16.55773
##     mort_5
## 1 26.61871
```

```r
# Target
ages <- c(20, 35, 45, 60, 75, 90)
targets <- data.frame(
  Age = ages,
  prev_mean = c(4.21, 7.19, 9.41, 12.03, 14.82, 18.59),
  prev_low  = c(3.88, 6.70, 8.85, 11.29, 13.64, 16.15),
  prev_high = c(4.61, 7.68, 10.07, 12.96, 15.94, 21.43),

  mort_mean = c(14.03, 13.26, 13.21, 13.68, 15.09, 23.87),
  mort_low  = c(10.80, 10.97, 10.58, 11.22, 11.80, 16.07),
  mort_high = c(17.32, 15.64, 16.00, 16.17, 18.44, 32.96)
)
head(targets, 2)
```

```
##   Age prev_mean prev_low prev_high mort_mean mort_low mort_high
## 1  20      4.21     3.88      4.61     14.03    10.80     17.32
## 2  35      7.19     6.70      7.68     13.26    10.97     15.64
```

```r
# ---------------
# Model predictions
# ---------------
# Prevalence
sim_prev <- data.frame(
  age = ages,
  value = as.numeric(data7e[1, grep("prev_", names(data7e))])
)

# Mortality
sim_mort <- data.frame(
  age = ages,
  value = as.numeric(data7e[1, grep("mort_", names(data7e))])
)

sim_prev
```

```
##   age     value
## 1  20  5.395328
## 2  35  7.813280
## 3  45  9.568771
## 4  60 12.389858
## 5  75 13.370230
## 6  90 15.922108
```

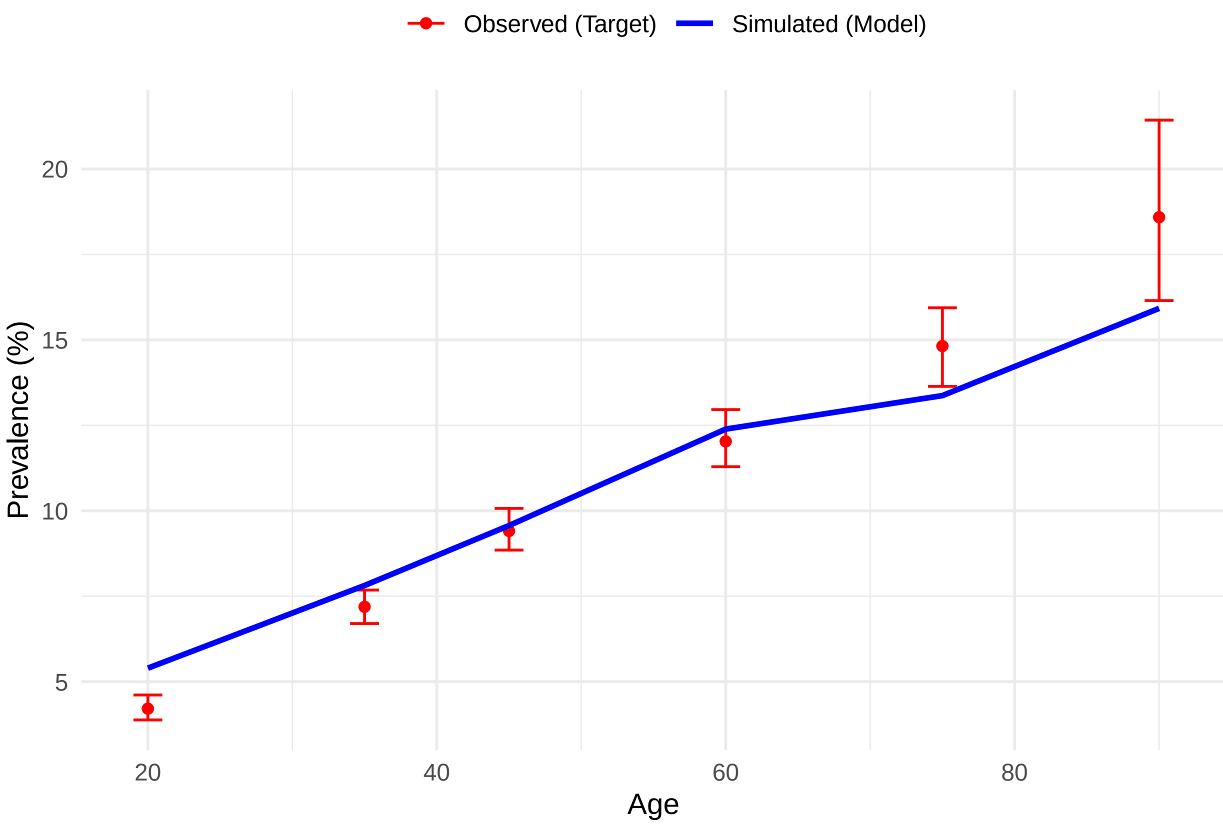
```r
sim_mort
```

```
##   age     value
## 1  20 10.772358
## 2  35 11.022364
## 3  45 12.518409
## 4  60  9.288824
## 5  75 16.557734
## 6  90 26.618705
```

```r
# ---------------
# Plot
# ---------------
```

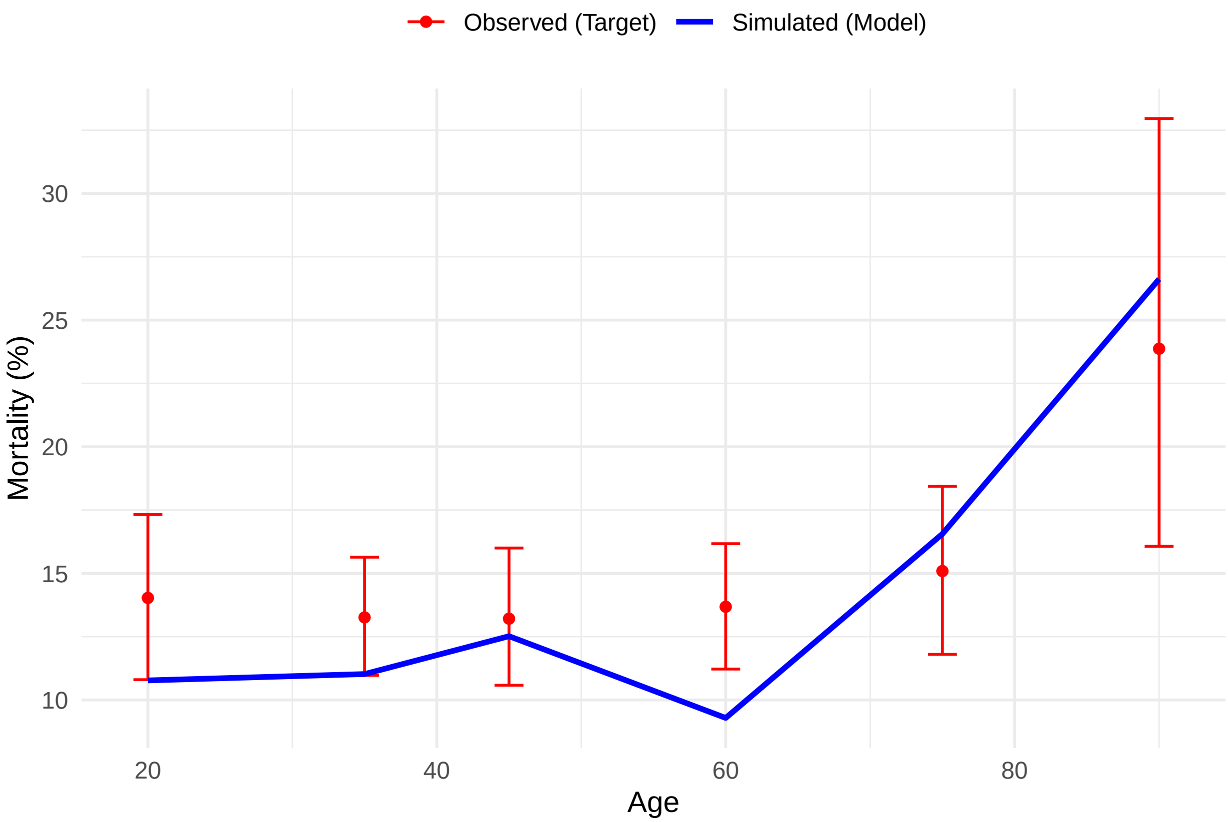
```r
# Plot: Prevalence
ggplot()+
  # Target
  geom_point(data = targets, aes(x = Age, y = prev_mean, colour = "Observed (Target)"))+
  geom_errorbar(data = targets,
                aes(x = Age, ymin = prev_low, ymax = prev_high, colour = "Observed (Target)"),
                width = 2)+
  # Simulated
  geom_line(data = sim_prev, aes(x = age, y = value, colour = "Simulated (Model)") , linewidth = 1)+

  scale_colour_manual(values = c("Observed (Target)" = "red",
                                 "Simulated (Model)" = "blue"))+
  labs(y = "Prevalence (%)",
       colour = " ")+
  theme_minimal()+
  theme(legend.position = "top")
```




```r
# Plot: Mortality
ggplot()+
  # Target
  geom_point(data = targets, aes(x = Age, y = mort_mean, colour = "Observed (Target)"))+
  geom_errorbar(data = targets,
                aes(x = Age, ymin = mort_low, ymax = mort_high, colour = "Observed (Target)"),
                width = 2)+
  # Simulated
  geom_line(data = sim_mort, aes(x = age, y = value, colour = "Simulated (Model)"), linewidth = 1)+
  scale_colour_manual(values = c("Observed (Target)" = "red",
                                 "Simulated (Model)" = "blue"))+
```

```r
  labs(y = "Mortality (%)",
       colour = " ")+
  theme_minimal()+
  theme(legend.position = "top")
```



7f

```r
# ---------------
# Load dataset
# ---------------
data7f <- read.csv("./rds203_problemset2/data/Q7f.csv")
head(data7e, 2)
```

```
##        beta0        beta1  recovery    mortDx   prev_0  prev_1   prev_2
## 1 0.01210111 0.0008811535 0.3573952 0.1462546 5.395328 7.81328 9.568771
##      prev_3   prev_4   prev_5   mort_0   mort_1   mort_2   mort_3   mort_4
## 1 12.38986 13.37023 15.92211 10.77236 11.02236 12.51841 9.288824 16.55773
##      mort_5
## 1 26.61871
```

```r
# ---------------
# Compute mean and range
# ---------------
# Prevalence
prev_mean <- colMeans(data7f[, grep("prev_", names(data7f))])
prev_min <- apply(data7f[, grep("prev_", names(data7f))], 2, min)
prev_max <- apply(data7f[, grep("prev_", names(data7f))], 2, max)
```

```r
# Mortality
mort_mean <- colMeans(data7f[, grep("mort_", names(data7f))])
mort_min <- apply(data7f[, grep("mort_", names(data7f))], 2, min)
mort_max <- apply(data7f[, grep("mort_", names(data7f))], 2, max)

# ---------------
# Create subsets
# ---------------
# Prevalence
sim_prev <- data.frame(
  age = ages,
  mean = prev_mean,
  min = prev_min,
  max = prev_max
)
rownames(sim_prev) <- NULL

# Mortality
sim_mort <- data.frame(
  age = ages,
  mean = mort_mean,
  min = mort_min,
  max = mort_max
)
rownames(sim_mort) <- NULL

sim_prev
```

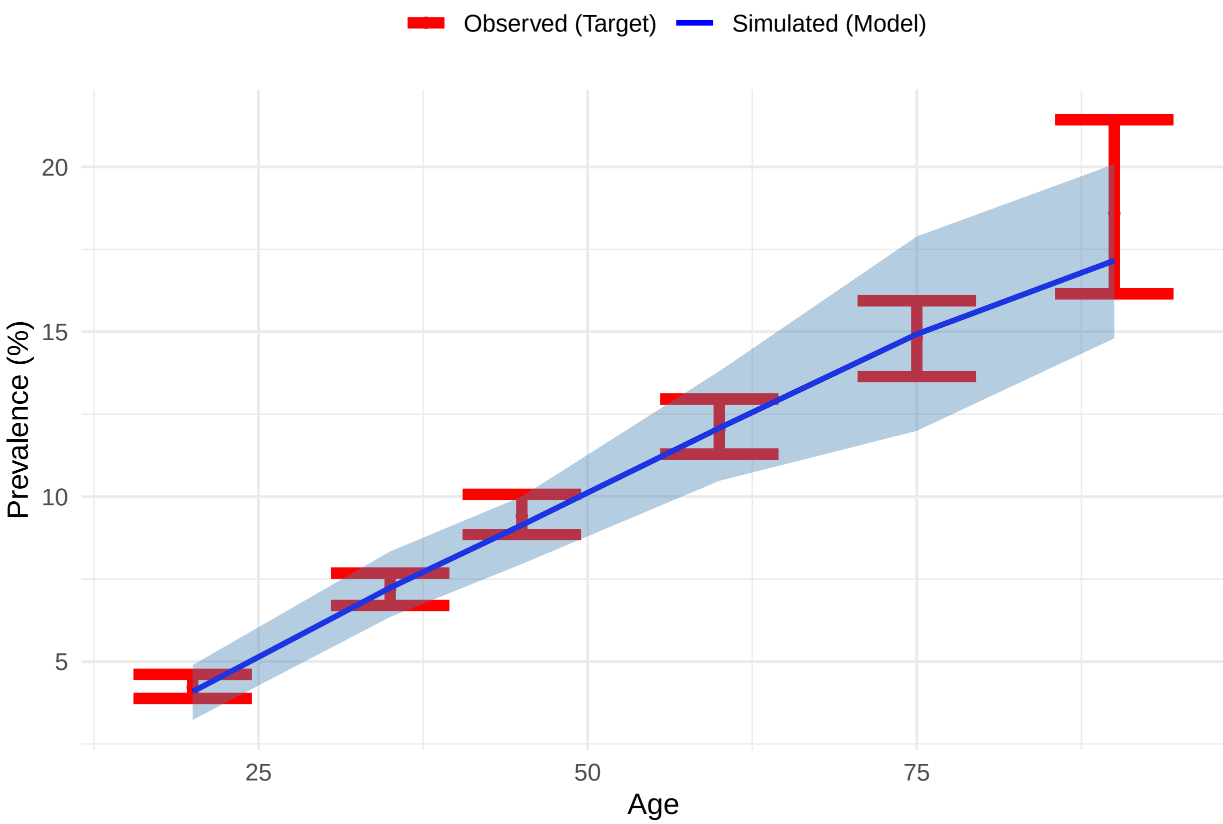
```
##   age      mean       min       max
## 1  20  4.086795  3.229233  4.891886
## 2  35  7.237290  6.349578  8.339585
## 3  45  9.137248  7.956482 10.004157
## 4  60 12.080499 10.477787 13.805310
## 5  75 14.927351 11.993671 17.890957
## 6  90 17.159063 14.800759 20.082531
```

```r
sim_mort
```

```
##   age     mean       min      max
## 1  20 14.29530  9.917355 19.03323
## 2  35 13.68259 10.702875 16.43110
## 3  45 12.98767 11.363636 15.67164
## 4  60 13.68251 10.380117 16.79873
## 5  75 16.75324 14.145383 20.04264
## 6  90 29.21366 23.648649 34.65347
```

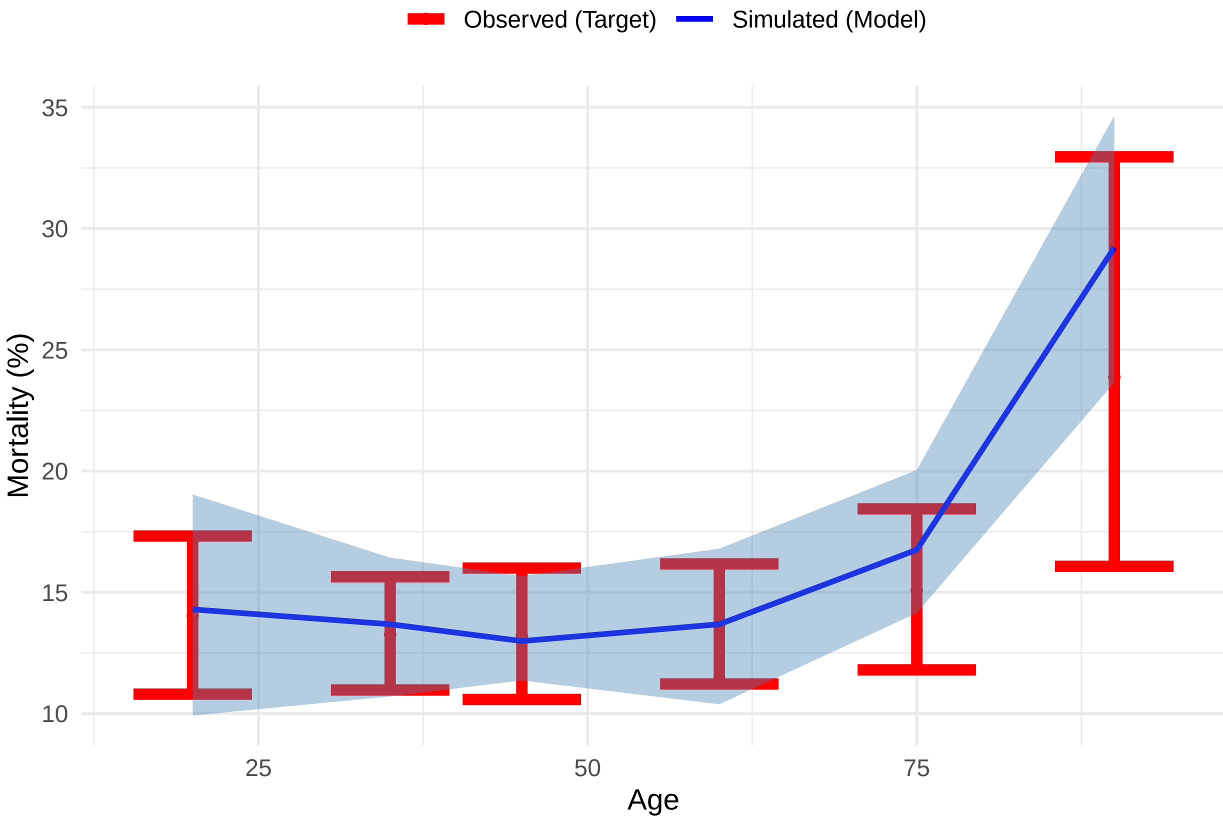
```r
# ---------------
# Plot
# ---------------
# Plot: Prevalence
ggplot()+
  # Target
  geom_point(data = targets, aes(x = Age, y = prev_mean, colour = "Observed (Target)"))+
  geom_errorbar(data = targets,
              aes(x = Age, ymin = prev_low, ymax = prev_high, colour = "Observed (Target)"),
```

```r
                linewidth = 2)+
  # Simulated
  geom_line(data = sim_prev, aes(x = age, y = mean, colour = "Simulated (Model)"), linewidth = 1)+
  geom_ribbon(data = sim_prev,
              aes(x = age, ymin = min, ymax = max),
              fill = "steelblue", alpha = 0.4)+
  scale_colour_manual(values = c("Observed (Target)" = "red",
                                 "Simulated (Model)" = "blue"))+
  labs(y = "Prevalence (%)",
       colour = " ")+
  theme_minimal()+
  theme(legend.position = "top")
```




```r
# Plot: Mortality
ggplot()+
  # Target
  geom_point(data = targets, aes(x = Age, y = mort_mean, colour = "Observed (Target)"))+
  geom_errorbar(data = targets,
                aes(x = Age, ymin = mort_low, ymax = mort_high, colour = "Observed (Target)"),
                linewidth = 2)+
  # Simulated
  geom_line(data = sim_mort, aes(x = age, y = mean, colour = "Simulated (Model)"), linewidth = 1)+
  geom_ribbon(data = sim_mort,
              aes(x = age, ymin = min, ymax = max),
              fill = "steelblue", alpha = 0.4)+
  scale_colour_manual(values = c("Observed (Target)" = "red",
                                 "Simulated (Model)" = "blue"))+
  labs(y = "Mortality (%)",
```

```r
         colour = " ")+
  theme_minimal()+
  theme(legend.position = "top")
```



Question 8

```r
data8 <- read.csv("./rds203_problemset2/data/Q8.csv")
head(data8, 2)
```

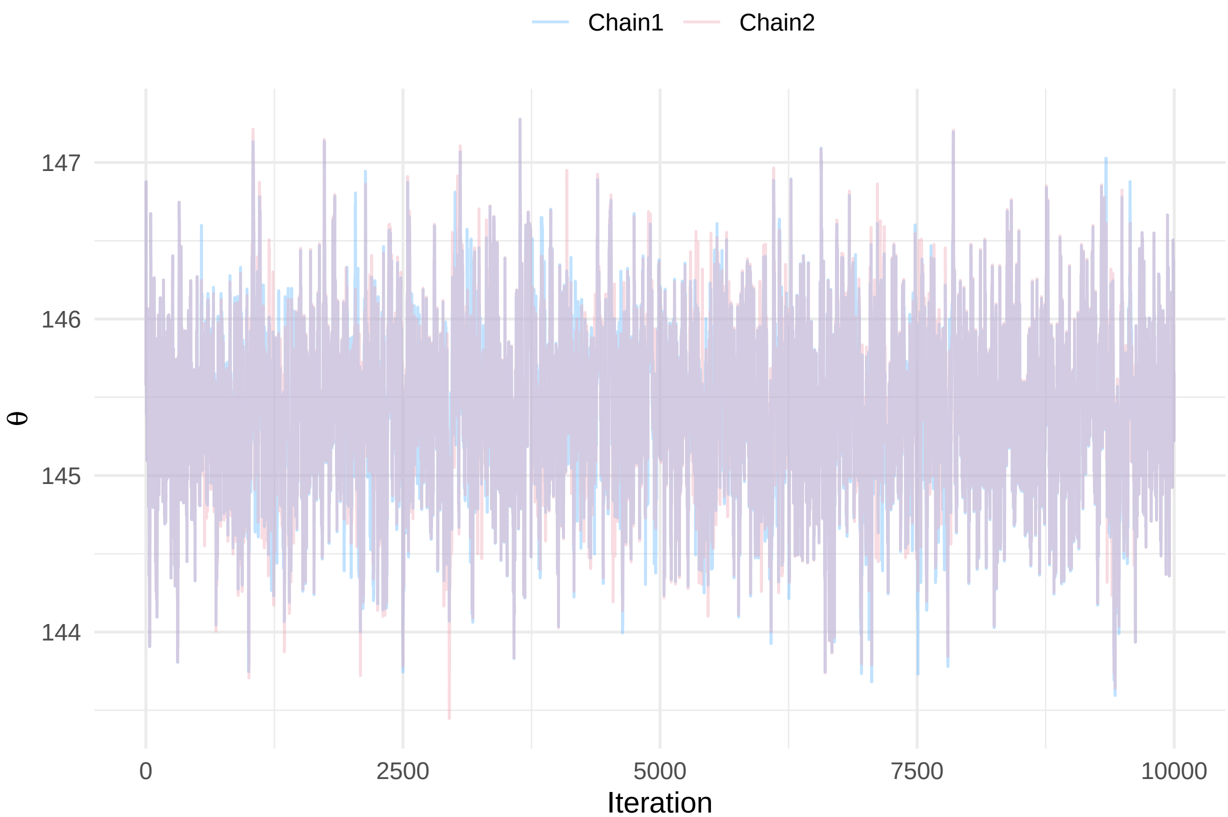
```
##   Iter   Chain1   Chain2
## 1    0 145.5759 145.5775
## 2    1 145.9843 145.9860
```

```r
data8_long <- data8|>
  pivot_longer(cols = c("Chain1", "Chain2"),
               names_to = "Chain",
               values_to = "theta")
head(data8_long, 2)
```

```
## # A tibble: 2 x 3
##    Iter Chain  theta
##   <int> <chr>  <dbl>
## 1     0 Chain1  146.
## 2     0 Chain2  146.
```

```r
# Plot tract of the chains
ggplot(data8_long, aes(x = Iter, y = theta, colour = Chain))+
  geom_line(alpha = 0.4)+
```

```
  scale_colour_manual(values = c(Chain1 = "steelblue1",
                                 Chain2 = "pink2"))+
  labs(x = "Iteration",
       y = expression(theta),
       colour = " ")+
  theme_minimal()+
  theme(legend.position = "top")
```

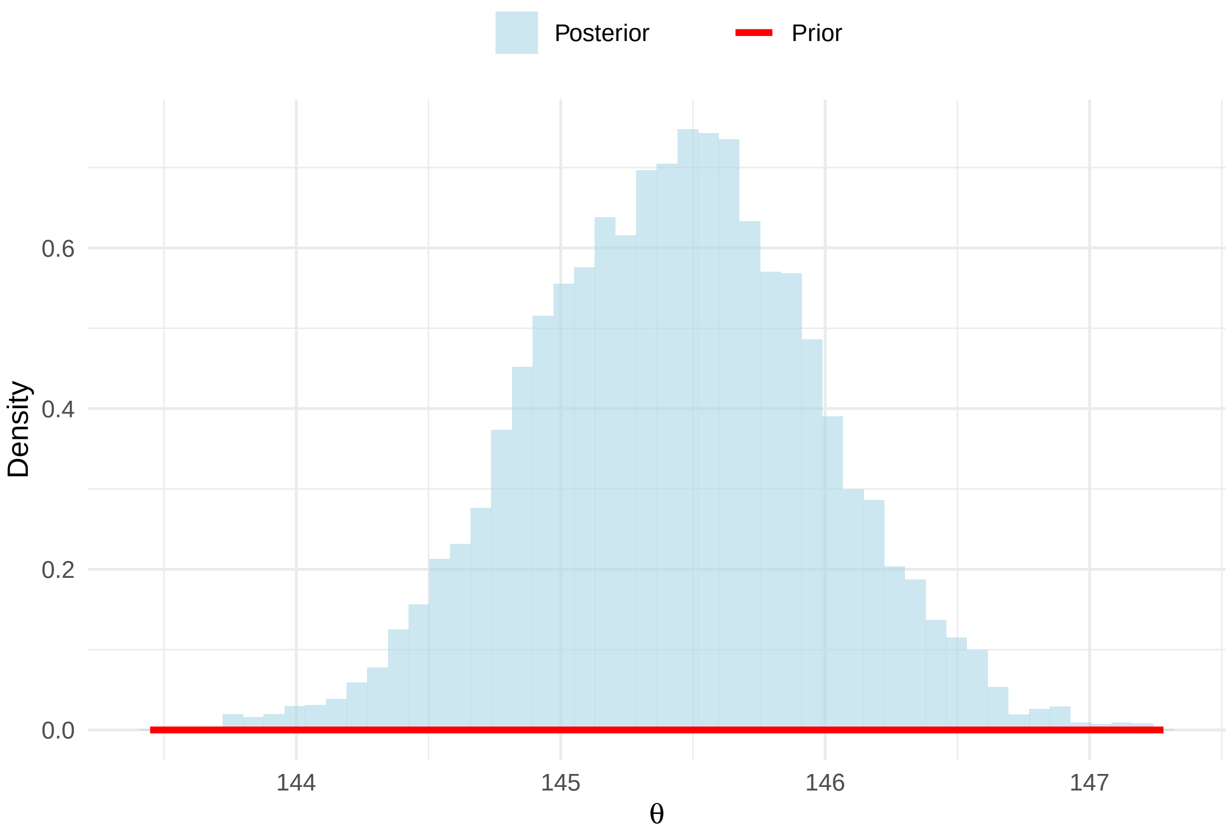



```
# Plot distribution of the posteriors vs prior
pos_df <- data.frame(theta = c(data8$Chain1, data8$Chain2))
head(pos_df, 2)
```

```
##      theta
## 1 145.5759
## 2 145.9843
```

```
ggplot(pos_df, aes(x = theta))+
  # Posterior
  geom_histogram(aes(y = after_stat(density), fill = "Posterior"),
                 bins = 50,
                 alpha = 0.6)+
  # Prior
  stat_function(aes(colour = "Prior"),
                fun = function(x) dnorm(x, 120, sqrt(2)),
                linewidth = 1.2)+
  scale_fill_manual(values = c("Posterior" = "lightblue"))+
  scale_colour_manual(values = c("Prior" = "red"))+
  labs(x = expression(theta),
       y = "Density",
```

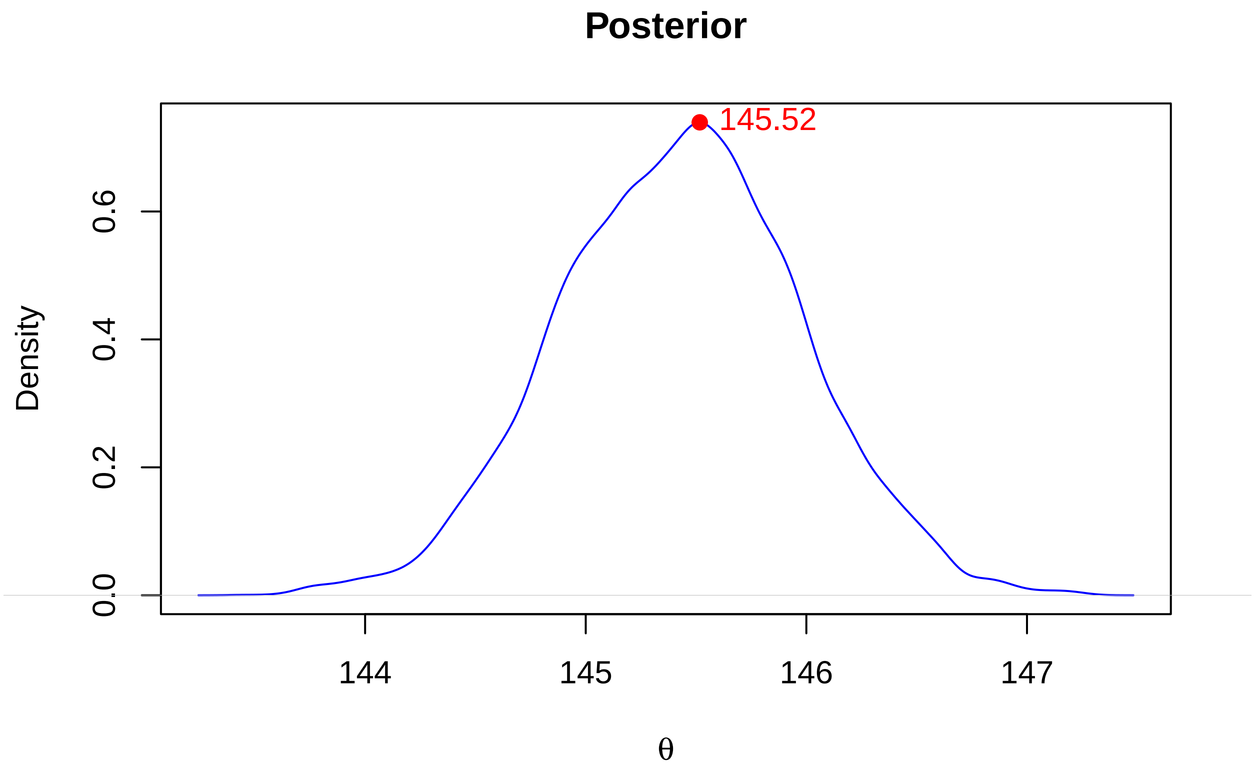
```
       colour = " ",
       fill = " ")+
  theme_minimal()+
  theme(legend.position = "top")
```



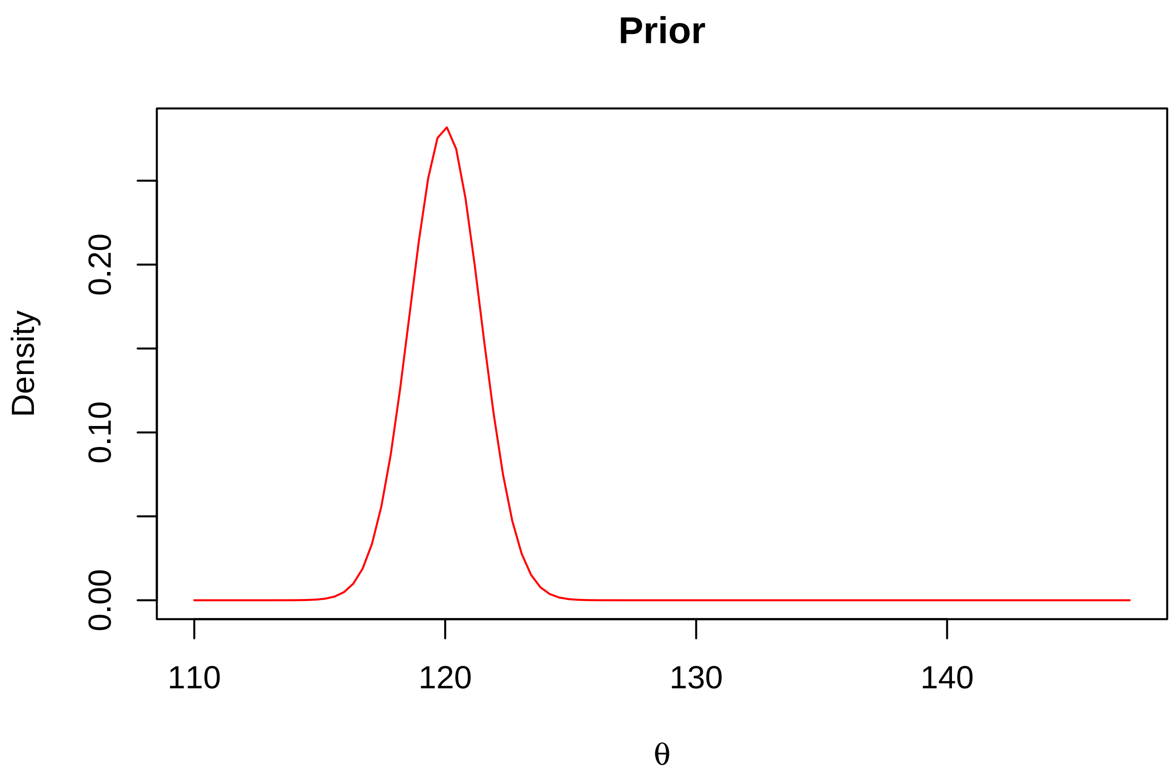

```
# Posterior
par(xpd = TRUE)

pos <- c(data8$Chain1, data8$Chain2)
plot(density(pos),
     col="blue",
     main = "Posterior",
     xlab = expression(theta),
     ylab = "Density")

# Add peak
d <- density(pos)
peak_x <- d$x[which.max(d$y)]
peak_y <- max(d$y)
points(peak_x, peak_y, col= "red", pch = 19)
text(peak_x, peak_y,
     labels = round(peak_x, 2),
     pos = 4, offset = 0.5, col = "red")
```




```r
# Prior
curve(dnorm(x, mean=120, sd=sqrt(2)),
      from=110, to=max(pos),
      col="red",
      main = "Prior",
      xlab = expression(theta),
      ylab = "Density")
```

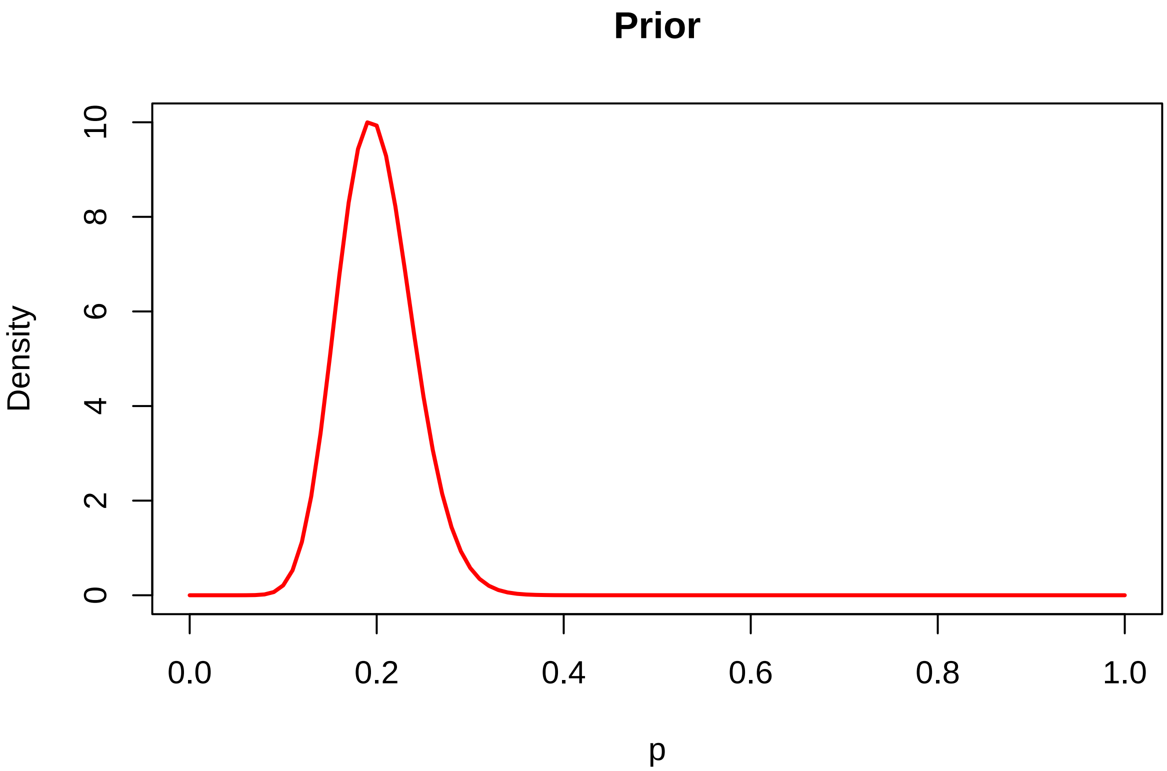


Question 9

```r
data9 <- read.csv("./rds203_problemset2/data/Q9.csv")
head(data9)
```

```
## [1] TransmissionProbability
## <0 rows> (or 0-length row.names)
```

```r
# Prior
curve(dbeta(x, 20, 80),
      from = 0, to = 1,
      col = "red",
      lwd = 2,
      main = "Prior",
      xlab = "p",
      ylab = "Density")
```


Prior
Density
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0
p